

Experimental Evaluation of a Multi-Layer Abstraction Approach for RL on Non-Markovian Tasks

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Outline

- ▶ Motivation
- ▶ Framework and Implementation
- ▶ Experimental Evaluation
- ▶ Conclusions



Motivation and Problem Statement

1 Motivation

Sparse Rewards

Informative feedback
is received only after
reaching the final goal.

Non-Markovian Task

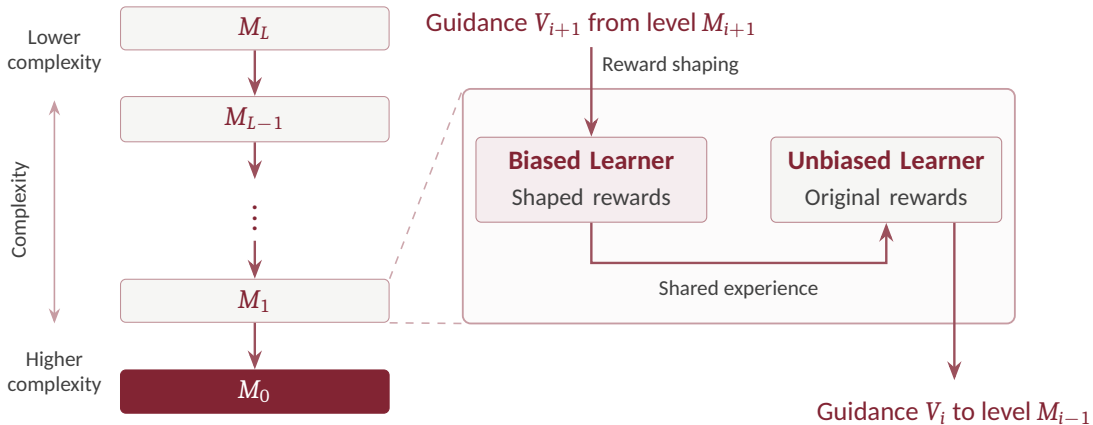
Task completion may depend
on the history of previously
observed events.

Prohibitive Learning



The Starting Point

2 Framework and Implementation

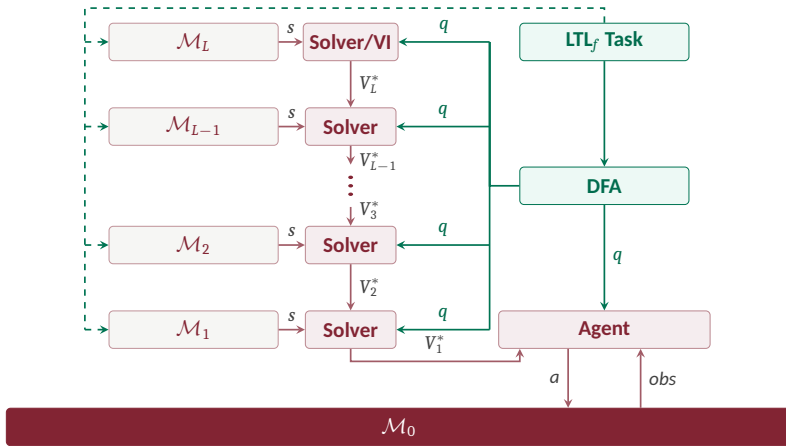


Framework by Cipollone et al. [1]



Extension Proposed in This Thesis

2 Framework and Implementation





Research Directions

3 Experimental Evaluation

RQ1: Direct Applicability

Can the original learning architecture be effectively applied to the considered ground domain?

RQ2: Scaling Temporal Complexity

How does abstraction-based guidance behave as temporal complexity increases?

RQ3: Learning Abstract Guidance

Can abstract guidance be learned and propagated across abstraction levels?

RQ4: Cyclic Temporal Tasks

Can abstraction-based guidance support recurring temporal behaviors?

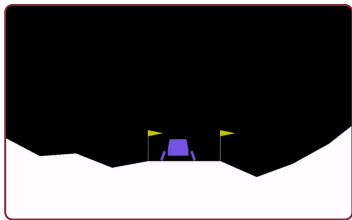


Experimental Setting

3 Experimental Evaluation

Ground Domain M_0

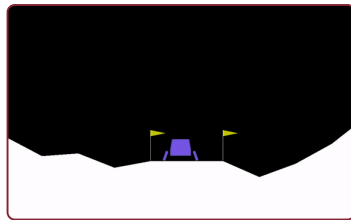
Gymnasium Lunar Lander



- $\mathcal{S} = (x, y, v_x, v_y, \theta, \omega, c_l, c_r)$
- $\mathcal{A} = \{\text{no-op, left-engine, right-engine, main-engine}\}$
- Original dense reward removed

Ground Domain M_0

Gymnasium Lunar Lander



- $\mathcal{S} = (x, y, v_x, v_y, \theta, \omega, c_l, c_r)$
- $\mathcal{A} = \{\text{no-op, left-engine, right-engine, main-engine}\}$
- Original dense reward removed



RQ1: Direct Applicability of the Framework

3 Experimental Evaluation

Learner interaction

- Biased learner exploits shaping
- Unbiased learner: no reliable benefit from biased experience

Biased / unbiased interaction diagram

Temporal misalignment

- Ground steps may leave abstract state unchanged
- $\gamma_s < 1$: artificial shaping penalties

Best success rate

$$\gamma_s = 0.99 \rightarrow 0\%$$

$$\gamma_s = 1 \rightarrow \approx 97\%$$

Practical choice: $\gamma_s = 1$.



RQ2: Increasing Temporal Complexity

3 Experimental Evaluation

Episodic task diagrams: $2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ phases

*Representative learning plot
Baseline vs. abstraction-based shaping*

- Baseline generally fails
- Shaping supports longer specifications
- Reliability decreases across training seeds
- Successful runs learn longer DFA prefixes

Longer temporal tasks mainly challenge learning reliability.



RQ3: Learning and Propagating Abstract Guidance

3 Experimental Evaluation

Coarse 90×90 grid $\longrightarrow$ fine 360×360 grid $\longrightarrow$ Lunar Lander

- Fine-grid Q-learning benefits from coarser guidance
- Multi-level learning when exact planning is impractical

Extracting the shaping potential

Value Iteration $\approx$ Policy Evaluation $>$ MaxQ
in shaping effectiveness

A successful learned policy does not automatically yield a good potential.



RQ4 and Additional Case Studies

3 Experimental Evaluation

Cyclic temporal tasks

*Horizontal / vertical
Triangular / rectangular patterns*

- Shaping enables repeated temporal-pattern satisfaction

Limits of the abstraction

- Spatial constraints: aligned predicates work well
- Landing tasks: missing physical information
- Position-only grids omit velocity, orientation, angular velocity, ground contacts

Guidance depends on preserving task-relevant information.



Conclusions and Future Work

4 Conclusions

Contributions

- Automated declarative LTLf tasks
- DFA across the hierarchy
- Configurable multi-layer implementation
- Complex-domain evaluation

Main findings

- Sparse temporal tasks become learnable
- Complexity affects reliability / exploration
- Value extraction is critical
- Task alignment limits shaping

Future work

- Task-aware exploration
- Richer physical abstractions
- Better value extraction / propagation
- General treatment of temporal misalignment

Abstraction-based guidance is effective when the abstraction preserves the information required to represent and support the temporal task.



Thank you for listening!
Any questions?



References

4 Conclusions

- [1] **Roberto Cipollone et al.** “Exploiting robot abstractions in episodic RL via reward shaping and heuristics”. In: *Robotics and Autonomous Systems* 193 (2025), p. 105116. DOI: 10.1016/j.robot.2025.105116.